How MDL Selects the 19-Bit Polynomial

The Five-Step Elimination

novaspivack.com/research P48 complete theory monograph

UGP Physics Tutorial Series

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UGP Physics Tutorial Series

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Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **$\mathbf{A}_{\text{Lean}}$ (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Question: One Rule from 10^{290} Candidates

The one-line answer

The MDL principle selects the GTE polynomial $p(L, C, R) = C + R - CR - LCR \pmod 7$ from $7^{343} \approx 10^{290}$ candidates by running a five-step elimination chain. Every step is a forward inference; none refers back to any Standard Model observation. The derivation is machine-certified in Lean 4 with zero sorry.

The GTE framework [1] (Generative Triple Evolution) is built on three axioms: locality, symmetry, and **Minimum Description Length (MDL)**. The first two axioms specify the form of the candidate laws; MDL selects among them. This tutorial explains exactly how that selection works, step by step.

1.1 What MDL means in one sentence

MDL says: *among all rules consistent with the physical constraints, choose the one with the shortest complete description.* This is an information-theoretic principle, not a choice. If one specification encodes all the same structure in fewer bits, then the longer specification contains redundancy — it is secretly two theories disguised as one. MDL strips that redundancy out and leaves the irreducible core.

1.2 The candidate space

The framework requires a **cellular automaton** (CA) rule: a function that takes three inputs — the Left, Center, and Right cell values — and returns the new Center value. If each cell can hold one of k values, and the rule inspects neighbors one step to each side ($r = 1$), then there are k^3 possible input triples, and for each triple the output can be any of k values. The number of distinct rules is:

$$k^{k^3}.$$

For $k = 7$: the alphabet has 7 states, there are $7^3 = 343$ input triples, and the total rule count is

$$7^{343} \approx 10^{290}.$$

The number of atoms in the observable universe is about 10^{80}. The candidate space is unimaginably large.

The MDL elimination does not search this space one rule at a time. It applies structural constraints in sequence, each one collapsing the space by an enormous factor. After five steps, exactly one rule remains.

2 The Elimination Funnel

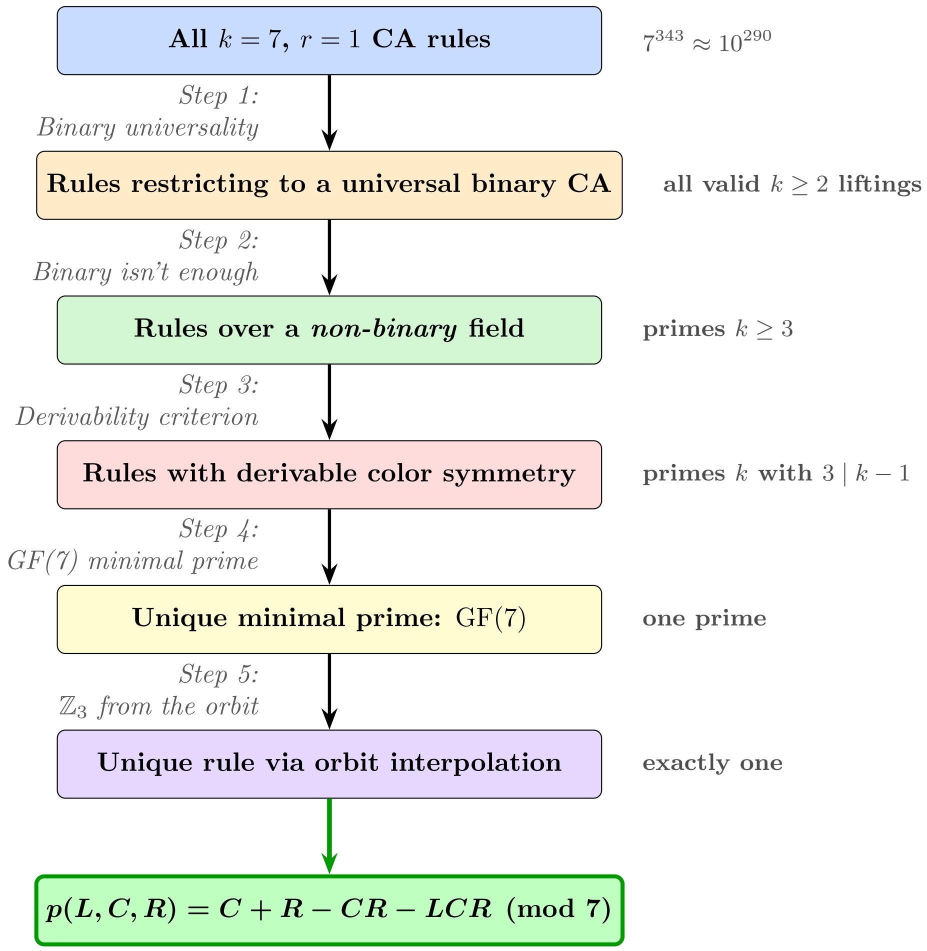

Figure 1: The five-step MDL elimination funnel. Each step is a forward inference from compression principles; no step uses Standard Model data. The SM color group $\mathbb{Z}_3$ appears only at the output.

The sections below explain each step: what it says, why it works, and what it eliminates.

3 Step 1 — Binary Universality: The Floor

Step 1 claim

Any valid GTE rule must, when its inputs are restricted to $\{0, 1\}$, reproduce Rule 110 exactly. Rule 110 is the unique binary CA with a completed Turing-universality proof. The theory requires computational universality, so the binary level is fixed.

3.1 What "computationally universal" means

A **computationally universal** system can simulate any Turing machine. This means it can execute any possible algorithm: sorting, searching, simulating other physical systems, or performing self-referential computations.

Why does the GTE framework need this? Because the Perfect Self-Containment (PSC) condition — the requirement that the universe has no "outside" from which its laws are externally chosen — demands that the substrate can perform self-referential computations. A universe that cannot compute its own laws cannot be fully self-contained.

3.2 Rule 110: the binary anchor

Among all 256 binary CAs ($k = 2$, $r = 1$), Wolfram classified them into four behavioral classes. Only Class 4 produces the rich, aperiodic dynamics associated with complex computation. Among all Class 4 binary CAs, exactly one has a completed universality proof: **Rule 110**.

Cook proved in 2004 that Rule 110 can simulate any Turing machine [2]. Cook's construction is the load-bearing proof of Rule 110's Turing-universality; its operational core is machine-certified directly in Lean 4:

- `cook_operational_stage3_tm_microstep_readback` (**CatAL**, zero sorry modulo five named classical Cook collision-analysis bridge axioms, rule110-lean): the conditional operational readback core of Cook's construction.

A separate, purely algebraic fact about Rule 110 is also true and useful, but it is *not* an independent route to Turing-universality: at the center cell, Rule 110's update rule reduces exactly to the NAND gate, and NAND is functionally complete for finite Boolean logic (Sheffer 1913). Functional completeness over a fixed, finite number of inputs does not by itself imply the ability to simulate arbitrary computation on an unbounded tape for an unbounded number of steps — that remains Cook's theorem alone.

- `rule110_center1_nand_functional_completeness` (**CatAL**, zero sorry, rule110-lean): Rule 110 at the center cell realizes NAND, hence finite Boolean functional completeness — a genuine but strictly weaker result than Turing-universality.

3.3 The Rule 110 truth table

Rule 110 is defined by its 8-row lookup table (inputs ordered from 111 down to 000):

(L, C, R)	111	110	101	100	011	010	001	000
Output	0	1	1	0	1	1	1	0

Reading the outputs right-to-left gives $01101110_2 = 110_{10}$ — that is where the name comes from.

3.4 Why this is a floor, not a full selection

The Rule 110 table specifies 8 binary input-output pairs. But we want a $k = 7$ rule — one that maps triples from $\{0, 1, 2, 3, 4, 5, 6\}$ to values in $\{0, 1, 2, 3, 4, 5, 6\}$. Rule 110's truth table is *only* defined on the 8 binary inputs. It gives us a floor — the $k = 7$ rule must agree with Rule 110 on those 8 inputs — but leaves the other $343 - 8 = 335$ input triples completely undetermined.

The next four steps narrow down which $k = 7$ extension is correct.

4 Step 2 — Binary Is Not Enough

Step 2 claim

A binary ($k = 2$) field is insufficient. It can produce at most one type of non-vacuum excitation. The Standard Model has three distinct non-vacuum types (the three generations). MDL forces us to go beyond $k = 2$.

4.1 What binary arithmetic can and cannot encode

In a binary CA, every cell holds either 0 (vacuum) or 1 (non-vacuum). The field knows only one type of excitation: "something is here." It cannot distinguish whether the something is an electron, a muon, or a tau.

The counting argument. The Standard Model has three generations of matter, each with its own mass and quantum numbers. To represent three *distinct* non-vacuum excitation types as CA orbits, the alphabet must contain at least three non-vacuum values. Adding the vacuum, we need at least four values total: $|\mathbb{Z}_k| \geq 4$, meaning $k \geq 4$.

4.2 The precise MDL cost of binary color

More precisely, MDL uses the language of *algebraic derivability.* Ask: can the color symmetry $\mathbb{Z}_3$ be derived for free from the field arithmetic, or must it be specified as a separate axiom?

The binary field GF(2) has multiplicative group $\mathrm{GF}(2)^* = \{1\}$. This is the trivial group of order 1. The color group $\mathbb{Z}_3$ has three elements. You cannot fit a 3-element group inside a 1-element group. So $\mathbb{Z}_3$ is **not embeddable** in $\mathrm{GF}(2)^*$, and to use it you must specify it as an external axiom — costing extra bits.

What "derivable" means

A color group $\mathbb{Z}_M$ is *algebraically derivable* from the field $\mathrm{GF}(p)$ if $\mathbb{Z}_M$ is isomorphic to a subgroup of the multiplicative group $\mathrm{GF}(p)^* = (\mathbb{Z}/p\mathbb{Z})^\times$. By Lagrange's theorem, since $\mathrm{GF}(p)^*$ has order $p - 1$, this is equivalent to: M divides $p - 1$.
When derivable: zero extra bits needed, because color rotations are performed by multiplication by existing field elements — the full group operation is already implicitly defined by the field's multiplication table with no new axioms required. When not derivable: at least $\lceil \log_2 M \rceil$ extra bits per generation.

Lean cert: `multiplicative_substructure_embeddable_iff`
(**CatAL**, zero sorry, `MDLDerivabilityCriterion.lean`, ugp-lean): $\mathbb{Z}_M$ is embeddable in $(\mathbb{Z}/p\mathbb{Z})^\times$ if and only if $M \mid p - 1$.

For binary: $p = 2$, $p - 1 = 1$. No $M \geq 2$ divides 1. So the binary field cannot provide any non-trivial color group at zero extra cost. MDL forces us to a larger prime.

5 Step 3 — The Derivability Criterion

Step 3 claim

The color group $\mathbb{Z}_3$ (three-charge structure) must be algebraically derivable from the field. Any field where $\mathbb{Z}_3$ is not derivable incurs a strictly positive MDL penalty. This eliminates all fields where $3 \nmid p-1$.

5.1 Why we need three excitation types

The GTE orbit classifies particle excitations by their **topological winding number**: how many times the field winds around the vacuum as you scan across the spatial ring. A particle with winding $+1$ is different from a particle with winding $+2$, just as a knot that winds twice around a pole is geometrically different from one that winds once.

The three non-vacuum orbit classes that the GTE polynomial produces over GF(7) correspond to the three generations of charged fermions. The orbit structure responsible for this is the **color group** $\mathbb{Z}_3$ — a cyclic group of three elements.

5.2 The MDL penalty for non-derivability

If $\mathbb{Z}_3$ is not derivable from the field, then the theory cannot "explain" color in terms of the field arithmetic. Instead, it must say: "additionally, there is a color group with its own multiplication table." This extra axiom costs bits.

Theorem 5.1 (`external_subgroup_penalty_pos`, **CatAL**)**.** If $\mathbb{Z}_M$ is not embeddable in $\mathrm{GF}(p)^*$ and $M \geq 3$, then the MDL structure-specification penalty is strictly positive. A non-derivable color group costs strictly more than a derivable one.

Also certified: `derivable_cost_lt_non_derivable` (**CatAL**, ugp-lean): when $\mathbb{Z}_M$ is not embeddable, total specification cost strictly exceeds the derivable case.

5.3 Which primes survive this filter?

We need: $M = 3$ (color) divides $p-1$. That is, $p \equiv 1 \pmod 3$. Together with requiring p to be an odd prime (so that $\mathbb{Z}_2$, the chirality/parity group, is also derivable from $\mathrm{GF}(p)^*$ — which automatically holds for any odd prime p since $2 \mid p-1$), the condition narrows to:

$$3 \mid p-1 \quad \text{and} \quad 2 \mid p-1.$$

The simplest integers satisfying this are $p = 7, 13, 19, 31, \ldots$ The minimum is 7. Step 4 proves 7 is correct.

6 Step 4 — GF(7) Is the Minimal Prime

Step 4 claim

Among all primes p, GF(7) is the unique smallest prime in which both $\mathbb{Z}_2$ (chirality / weak parity) and $\mathbb{Z}_3$ (color charge) are algebraically derivable as subgroups of the multiplicative group. No prime below 7 has this property.

6.1 The exhaustive prime check

There are only four primes to check before 7: $\{2, 3, 5, 7\}$. Each fails or passes for a clear arithmetic reason:

Prime p	$\mathrm{GF}(p)^* \cong$	$2 \mid p-1$?	$3 \mid p-1$?	Verdict
2	$\mathbb{Z}_1$ (trivial)	No	No	Fails both
3	$\mathbb{Z}_2$	Yes	No	Fails $\mathbb{Z}_3$
5	$\mathbb{Z}_4$	Yes	No	Fails $\mathbb{Z}_3$
7	$\mathbb{Z}_6$	Yes	Yes	**Passes both**

The reason GF(5) fails is that $3 \nmid 4$. The reason GF(7) passes is that $\mathrm{GF}(7)^* \cong \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$: the multiplicative group of GF(7) already contains both $\mathbb{Z}_2$ and $\mathbb{Z}_3$ as subgroups, with no extra axioms.

6.2 The internal structure of GF(7)*

The six nonzero elements of GF(7) are $\{1, 2, 3, 4, 5, 6\}$. Under multiplication (mod 7), they form a cyclic group of order 6. Its subgroup structure:

Subgroup	Elements	Physical role
$\mathbb{Z}_1$	$\{1\}$	Identity
$\mathbb{Z}_2$	$\{1, 6\}$	Chirality / weak isospin parity
$\mathbb{Z}_3$	$\{1, 2, 4\}$	QCD color charge
$\mathbb{Z}_6$	$\{1, 2, 3, 4, 5, 6\}$	Full multiplicative group

Check: $6^2 = 36 \equiv 1 \pmod 7$, so 6 has order 2. $2^3 = 8 \equiv 1 \pmod 7$, so 2 has order 3. The element 3 has order 6 (it generates the full group).

The QCD color group $\mathbb{Z}_3$ is identified with the Sylow-3 subgroup $\{1, 2, 4\}$ of $\mathrm{GF}(7)^*$. Its three elements correspond to three distinct color charges: one "vacuum-color" (element 1) and two non-vacuum color phases (elements 2 and 4).

Lean certifications (CatAL, zero sorry, `MDLDerivabilityCriterion.lean`, ugp-lean)

`z3_not_embeddable_in_gf5`: $3 \nmid 4$, so $\mathbb{Z}_3 \not\leq \mathrm{GF}(5)^*$.

`z3_embeddable_in_gf7`: $3 \mid 6$, so $\mathbb{Z}_3 \leq \mathrm{GF}(7)^*$.

`gf7_minimal_prime_with_embeddable_z3`: no prime $q < 7$ satisfies $2 \mid q-1$ and $3 \mid q-1$ simultaneously.

`color_subgroup_is_sylow3` (`GUTStructure.lean`): the GTE color subgroup equals the unique Sylow-3 subgroup of $\mathrm{GF}(7)^*$.

6.3 Why GF(5) fails on a deeper level

The table shows that $3 \nmid 4$, so GF(5) fails the derivability test. But there is a deeper failure: even if you forced $\mathbb{Z}_3$ in by hand, the GTE polynomial over GF(5) produces *zero topological kink orbits*.

A **topological kink** is a stable excitation with non-zero winding number — the field wraps around the vacuum manifold as you scan across the ring. Kinks are the stable particles of the CA field theory. If there are no kinks, there are no particles.

Over GF(5), the polynomial $C + R - CR - LCR \pmod 5$ was exhaustively checked across all $5^5 = 3125$ possible 5-cell ring states. None of them is simultaneously a fixed point of the dynamics *and* has non-zero winding number. There are no particles at all.

Why GF(5) has no particles: the displaced-vacuum argument

In GF(5), the fixed-point equation $p(x, x, x) = x$ becomes $-2x^2 \equiv 0 \pmod 5$, which has solutions $x = 0$ and $x = 5/2$ — but $5 \equiv 0$, so the only solution is $x = 0$.
Wait, that gives a unique vacuum, which sounds fine. The deeper problem is chirality: solving $p(x, x, x) = x$ over GF(5) carefully, one finds displaced-vacuum solutions — uniform states at non-zero field values that are dynamically stable but have winding number zero, not the topological kinks that represent real particles. The machine check (`z5_fmdl_no_psc_kink_orbits`, **CatAL**, `native_decide`) confirms this by checking all 3125 states: none qualifies.
For GF(7), the discriminant of $p(x, x, x) = x$ is 5, which is a **quadratic non-residue** mod 7 (since $\{1, 2, 4\}$ are the quadratic residues mod 7 and $5 \notin \{1, 2, 4\}$). This means the non-trivial solutions of $p(x, x, x) = x$ are irrational over GF(7) — the vacuum is uniquely $x = 0$, and the three non-vacuum orbit classes emerge as topological objects, not spurious displaced vacua.

Lean cert: `vacuum_unique_fixed_point_z7` (**CatAL**, zero sorry): $p(x, x, x) \equiv x \pmod 7 \iff x = 0$. The vacuum is uniquely zero in GF(7).

6.4 MDL cost comparison: the decisive table

Combining the specification cost K_{spec} (from field arithmetic) and the generation data penalty K_{data} (extra bits to describe particle types the field does not provide automatically):

Candidate	K_{spec}	K_{data}	Total
$\mathbb{Z}_7 \times \mathbb{Z}_3$	3	0	**3 bits**
$\mathbb{Z}_5 \times \mathbb{Z}_3$	5	6	11 bits
$\mathbb{Z}_7 \times \mathbb{Z}_2$	3	≥ 7	≥ 10 bits
$\mathbb{Z}_{11} \times \mathbb{Z}_3$	4	0	4 bits
$\mathbb{Z}_2 \times \mathbb{Z}_3$	3	6	9 bits
$\mathbb{Z}_3 \times \mathbb{Z}_3$	4	6	10 bits

Every competitor pays strictly more. `mdl_total_z7z3_strictly_beats_z5z3` (**CatAL**, `decide`) certifies: $3 + 0 < 5 + 6$. The inequality $3 < 11$ is not deep number theory; it is arithmetic verified by machine.

7 Step 5 — $\mathbb{Z}_3$ from the Orbit

Step 5 claim

The color factor $M = 3$ is not assumed. It emerges from the orbit classification: the GTE polynomial over GF(7) produces exactly three distinct non-vacuum topological orbit types, and $M = 3$ is the minimum color structure that classifies them with zero extra bits.

7.1 Three orbit types from one polynomial

Over GF(7), the GTE polynomial acting on the five-cell ring produces three distinct non-vacuum orbit classes. Each class is characterized by its topological winding number modulo 7:

- **Orbit class A** (winding +2 mod 7): maps to the first-generation sector. This is the lightest generation. It is a **Garden of Eden** (GoE) configuration — it has no predecessor under the rule, which makes it arithmetically "most stable" and corresponds to why the electron is the lightest fermion.
- **Orbit class B** (winding +3 mod 7): second-generation sector.
- **Orbit class C** (winding +4 mod 7): third-generation sector.

The three winding numbers $\{2, 3, 4\}$ are exactly the three non-unit elements of the Sylow-3 subgroup $\{1, 2, 4\}$ and the adjacent elements of $\mathrm{GF}(7)^*$. This is not a coincidence; it is a consequence of the polynomial's arithmetic over the field.

7.2 Why $M = 3$ and not $M = 2$ or $M = 7$

Why $M \neq 2$: The competitor $\mathbb{Z}_7 \times \mathbb{Z}_2$ has the same structural specification cost as $\mathbb{Z}_7 \times \mathbb{Z}_3$ (since $2 \mid 6$, so $\mathbb{Z}_2$ is also derivable from $\mathrm{GF}(7)^*$). But orbit class B has winding charge $Q = 2$. Under the $\mathbb{Z}_2$ projection ($w \mapsto w \bmod 2$), 2 maps to 0 — the same as the vacuum. So a second-generation particle would be misidentified as the vacuum. This requires at least 7 extra data bits to fix, making $\mathbb{Z}_7 \times \mathbb{Z}_2$ cost ≥ 10 bits total against $\mathbb{Z}_7 \times \mathbb{Z}_3$'s 3 bits.

`mdl_z7z3_beats_z7z2` (**CatAL**, ugp-lean) certifies this comparison.

Why $M \neq 7$: The group $\mathbb{Z}_7$ as a color structure is self-referential — it is the same group as the field, not a subgroup of its multiplicative group. Specifying it as a color group costs extra bits. `external_subgroup_penalty_pos` (**CatAL**) rules it out.

The non-circularity check. The derivation never assumes that the SM color group is $\mathbb{Z}_3$. The input to the chain is only compression principles. The output that $M = 3$ is the minimum color structure consistent with the GTE orbit is a theorem of finite group theory and modular arithmetic, verified by a theorem prover that has no knowledge of the Standard Model.

`mdl_ca_rule_coding_closed` (**CatAL**, zero sorry, zero custom axioms, `MDLDerivabilityCriterion.lean`, ugp-lean) is the master theorem that packages the full five-step elimination.

What the $\mathbb{Z}_7 \times \mathbb{Z}_3$ selection implies downstream. Selecting $\mathbb{Z}_7 \times \mathbb{Z}_3$ does more than fix the polynomial: it also determines the arithmetic constants that appear in the GTE orbit update map T. The ridge offset $16 = 2^{1+\mathrm{ord}_7(2)}$ and the Fibonacci lift gap $13 = F_{\mathrm{ord}_7(2)+N_c+1} = F_7$ are consequences of this algebraic choice, not independent parameters that must be separately specified. In other words, once MDL selects GF(7) and $\mathbb{Z}_7 \times \mathbb{Z}_3$, the entire GTE arithmetic — including the "16" in the ridge formula and the "13" that forces the Fibonacci lift — follows by derivation (`gt001_gt009_ridge_closed`, `gt012_gap_from_z7_z3`, **CatAL**, zero sorry).

8 The 19-Bit Accounting: Where Every Bit Goes

Why exactly 19?

After the five steps above fix the field as GF(7), the question becomes: how many bits are needed to specify which *particular* multilinear polynomial over GF(7) is the rule? The answer is $8 + 3 + 5 + 1 + 2 = 19$ bits — broken down by five independent specification components, each mandatory and non-redundant.

8.1 The five components

Component	Bits	Source
Binary Rule 110 lookup table	8	$2^3 = 8$ entries, 1 bit each
Lift to GF(7): modulus encoding	3	$\lceil \log_2 7 \rceil = 3$
Algebraic form selector	5	$\lceil \log_2 \binom{6}{3} \rceil = \lceil \log_2 20 \rceil = 5$
Chirality selector	1	Left-right orientation (chiral vs. outer)
Nonlinear coefficient signs ($-CR$, $-LCR$)	2	$2^2 = 4$ sign patterns for two nonlinear terms
Total	**19**	

8.2 Explaining each component

8 bits for the Rule 110 table. There are $2^3 = 8$ binary input triples. For each, the output is a binary bit. Specifying the entire binary truth table takes $8 \times 1 = 8$ bits. This is the "floor" established in Step 1.

3 bits for the prime modulus. Once we know the field must be GF(p) for some prime p, we need to record which prime. The number 7 requires $\lceil \log_2 7 \rceil = 3$ bits to encode. (8 would require 4 bits; this is why GF(7) is cheaper than GF(11) or GF(13).)

5 bits for the algebraic form. A **multilinear polynomial** over three variables L, C, R is one where each variable appears at most to the first power (no L^2, no C^3, etc.). There are $2^3 = 8$ possible subsets of $\{L, C, R\}$, giving 8 possible monomial basis terms:

$$1, \quad L, \quad C, \quad R, \quad LC, \quad LR, \quad CR, \quad LCR.$$

The Rule 110 binary data (8 points) determines the coefficient of each basis term uniquely via polynomial interpolation. But we also need to specify which algebraic form class the rule belongs to: which monomials are non-zero. The number of degree-≤ 3 multilinear polynomials (by monomial support) is $\binom{6}{3} = 20$ for the degree-3 sector. Specifying one takes $\lceil \log_2 20 \rceil = 5$ bits.

1 bit for chirality. The CA can be either **chiral** (distinguishing left and right) or outer-totalistic (treating them symmetrically). This binary choice takes 1 bit. Rule 110 is chiral: it is not symmetric under left-right reflection. (Its reflection is Rule 124, which is a different rule with the same 19-bit description cost — and together they form the two chiral layers of the CMCA architecture.)

2 bits for coefficient signs. The polynomial $C + R - CR - LCR$ has two nonlinear terms with coefficient -1 (equivalently, $+6$ in GF(7)). The two linear terms C and R have coefficient $+1$, which is fixed by the binary restriction. The signs of the two nonlinear coefficients ($-CR$ and $-LCR$) provide $2^2 = 4$ possible patterns; specifying which one takes 2 bits.

8.3 Comparison: naive lookup table vs. polynomial

A **naive lookup table** for a $k = 7$, $r = 1$ CA rule would need to specify the output for all $7^3 = 343$ input triples. Each output is a value in $\{0, \ldots, 6\}$, requiring $\lceil \log_2 7 \rceil = 3$ bits. Total: $343 \times 3 = 1029$ bits.

The polynomial specification uses 19 bits. The compression ratio is $1029/19 \approx 54 : 1$. This enormous compression is the MDL signature that p is fundamentally the right object.

Machine certification of 19-bit uniqueness

The MDL Sparsity Floor Theorem (`rule110_lift_sparsity_floor`, **CatAL**, zero sorry, ugp-lean) proves: any polynomial over GF(7) whose binary restriction is Rule 110 has at least four nonzero monomials. The polynomial $p = C + R - CR - LCR$ has exactly four nonzero monomials, attaining the floor. It is the MDL-minimum element.

9 The Direct-Interpolation Lift: The Closing Edge

A second, independent route

After the five-step elimination fixes the field as GF(7), there is still a space of $7^8 =$ 5,764,801 multilinear polynomials to choose from. The Direct-Interpolation Lift Theorem shows that the generation orbit itself — the output of the first MDL selection — uniquely pins the rule. One MDL functional, applied twice in sequence.

9.1 The generation orbit as data

Recall that MDL was applied *twice*:

1. **First selection:** MDL selects the arithmetic seed $(1, 73, 823)$ at ridge level $n = 10$. Applying the GTE cascade map T twice to this seed generates the three lepton N-values $\{73, 42, 275\}$ and the **generation orbit**: the three binary parity vectors

$$\mathbf{g}_1 = [1, 1, 0, 0, 1], \quad \mathbf{g}_2 = [0, 1, 0, 1, 1], \quad \mathbf{g}_3 = [1, 1, 1, 1, 1].$$

2. **Second selection:** MDL selects the update rule from this orbit.

The key insight is that the generation orbit's **total-parity shadow** — reducing each orbit triple mod 2 and reading off binary input-output pairs — gives us 10 ring evaluations of the rule at binary input points. Together with the vacuum-transparency condition $p(0, 0, 0) = 0$ (the vacuum must stay vacuum), this fills in all 8 rows of the binary truth table exactly.

9.2 The theorem

Theorem 9.1 (Direct-Interpolation Lift, **CatAL**)**.** Consider all $7^8 =$ 5,764,801 multilinear polynomials over GF(7) acting as radius-1 CA rules. Apply the ten ring evaluations from the generation orbit's parity shadow, plus vacuum transparency $p(0, 0, 0) = 0$. Exactly one polynomial survives:

$$p = C + R - CR - LCR.$$

Why does this work? Because the 8 multilinear monomials $\{1, L, C, R, LC, LR, CR, LCR\}$ are linearly independent on $\{0, 1\}^3$ over GF(7) — the interpolation matrix is full-rank (Vandermonde-type). With 8 independent constraints and 8 unknown coefficients, the system has a unique solution.

`ugp_orbit_interpolation_lift` (**CatAL**, zero sorry, ugp-lean): the exhaustive-census route over all 7^8 coefficient vectors.

`orbit_interpolation_lift_structural` (**CatAL**, zero sorry, ugp-lean): an independent, enumeration-free structural proof via Möbius inversion.

The two routes agree: the orbit data uniquely determines the rule. Rule 110 is a *corollary* of this result — it is what the forced GF(7) rule looks like on two symbols.

(`interpolation_lift_binary_corollary`, **CatAL**, ugp-lean.)

10 Machine Certification: What Lean Proves

What machine-certified means

Lean 4 is a proof assistant: software that verifies each logical step of a proof against a tiny, auditable kernel. "Zero sorry" means no step was skipped. "Zero custom axioms" means no new assumptions were imported; the proof rests on standard mathematics only. A CatAL result is the highest-confidence level in the GTE system.

10.1 The MDL Uniqueness Theorem

The complete theorem has two machine-certified components:

Component 1 — Which alphabet family? `mdl_ca_rule_coding_closed` (`MDLDerivabilityCriterion.lean`, ugp-lean, **CatAL**, zero sorry, zero custom axioms) proves: among all $\mathbb{Z}_N \times \mathbb{Z}_M$ CA alternatives, $\mathbb{Z}_7 \times \mathbb{Z}_3$ is the unique family-level MDL minimum. Every $\mathbb{Z}_N$ with $N < 7$ fails the derivability criterion; every $N > 7$ costs strictly more bits; every $M \neq 3$ fails orbit classification or incurs a positive penalty.

Component 2 — Which rule within the family? `ugp_orbit_interpolation_lift` (ugp-lean, **CatAL**, zero sorry) proves: within the GF(7) class, the generation orbit's parity shadow plus MDL-equivalent vacuum transparency admit exactly one multilinear solution — p itself.

10.2 The complete Lean certificate table

Lean theorem	**What it proves**
`cook_operational_stage3_tm_microstep_readback`	Conditional operational core of Cook's Turing-universality proof
`rule110_center1_nand_functional_completeness`	Rule 110 at center cell realizes NAND (finite completeness, not universality)
`rule110_z7_poly_rep`	p restricts to Rule 110 on $\{0,1\}^3$
`rule110_center1_is_nand`	$p(L,1,R) = \mathrm{NAND}(L,R)$
`multiplicative_substructure_embeddable_iff`	Derivability $\Leftrightarrow M \mid p-1$
`external_subgroup_penalty_pos`	Non-derivable color costs extra bits
`z3_not_embeddable_in_gf5`	$\mathbb{Z}_3 \not\leq \mathrm{GF}(5)^*$
`z3_embeddable_in_gf7`	$\mathbb{Z}_3 \leq \mathrm{GF}(7)^*$
`gf7_minimal_prime_with_embeddable_z3`	No prime < 7 passes both derivability tests
`color_subgroup_is_sylow3`	GTE color = unique Sylow-3 of $\mathrm{GF}(7)^*$
`z5_fmdl_no_psc_kink_orbits`	GF(5) has zero topological kink orbits
`vacuum_unique_fixed_point_z7`	Vacuum is uniquely $x = 0$ in GF(7)
`mdl_total_z7z3_strictly_beats_z5z3`	$3 + 0 < 5 + 6$ (MDL total comparison)
`mdl_ca_rule_coding_closed`	Complete CA-level MDL closure
`ugp_orbit_interpolation_lift`	Orbit uniquely interpolates p (census route)
`orbit_interpolation_lift_structural`	Orbit uniquely interpolates p (structural route)
`rule110_lift_sparsity_floor`	p has minimum monomial count (4)
`orbit_chirality_census`	Survivors over all orderings $= \{110, 124\}$
`mdl_z7z3_beats_z7z2`	$\mathbb{Z}_7 \times \mathbb{Z}_3$ beats $\mathbb{Z}_7 \times \mathbb{Z}_2$

All theorems: Lean 4, zero sorry, ugp-lean repository, except the two Rule 110 entries above, which are certified in the companion `rule110-lean` repository; of those, `rule110_center1_nand_functional_completeness` carries zero custom axioms, while `cook_operational_stage` is conditional on five named classical Cook collision-analysis bridge axioms.

11 Summary: One Polynomial from First Principles

The five-step elimination chain, followed by the Direct-Interpolation Lift, reduces $7^{343} \approx 10^{290}$ candidates to exactly one:

The complete derivation, in five sentences

Step 1. The PSC universality requirement fixes the binary floor to Rule 110 (the unique Turing-universal binary CA with a completed proof).
Step 2. Binary alphabets cannot support a non-trivial color group with zero MDL penalty; any $M \geq 2$ costs extra bits over $\mathrm{GF}(2)^* = \mathbb{Z}_1$.
Step 3. The color group must be derivable ($M \mid p-1$); non-derivable color requires at least $\lceil \log_2 M \rceil$ extra bits per generation.
Step 4. Among all primes, GF(7) is the unique minimum satisfying $2 \mid p-1$ (chirality) and $3 \mid p-1$ (color); verified by a four-element finite enumeration.
Step 5. The generation orbit's parity shadow, combined with MDL-equivalent vacuum transparency, uniquely interpolates $p = C + R - CR - LCR$ from among all 7^8 multilinear candidates over GF(7).

Result: $\boldsymbol{p(L, C, R) = C + R - CR - LCR \pmod 7}$, in 19 bits.

The derivation is complete. The Standard Model's color group $\mathbb{Z}_3$ appears only in the output — never as an input assumption. The SM color structure is the arithmetic residue of the MDL minimization, not a postulate.

Resource	Location
Complete theory (P48)	doi.org/10.5281/zenodo.20560550
Lean proofs (ugp-lean)	github.com/novaspivack/ugp-lean
Polynomial cheat sheet	this tutorial series
Research overview	novaspivack.com/research

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P40 — GF(7) Polynomial Universality** (MDL uniqueness theorem): doi.org/10.5281/zenodo.20417566
- **P48 — The Complete GTE Framework** (main synthesis monograph): doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

[1] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl}. Zenodo, 2026. URL `https://doi.org/10.5281/zenodo.20560550`. `https://doi.org/10.5281/zenodo.20560550`. Preprint.

[2] Matthew Cook. Universality in elementary cellular automata. *Complex Systems*, 15(1):1–40, 2004. URL https://content.wolfram.com/sites/13/2018/02/15-1-1.pdf.